HAPPIEST HEALTH DIGITAL TWIN CHALLENGE

HimanshuDigitalTwin

A patient-specific view of what blood glucose may do next.

30/60/120

minute glucose forecast horizons

Connects patient history, glucose readings, meals and wearable observations. Withholds forecasts when the sensor signal is unreliable.

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Research demonstrator

A glucose reading is a snapshot. The next two hours are not.

Meals, activity, patient phenotype and sensor quality can change what happens after the same starting value.

Synthetic patient example · 12:15 PM · 144 mg/dL

Recent carbohydrate context: 59.7 g · Wearable heart rate: 82 bpm · Illustrative replay, not a calibrated clinical forecast

NOW	+30 MIN	+60 MIN	+120 MIN
144	**155.5**	**156.0**	**145.4**
mg/dL observed	mg/dL forecast	mg/dL forecast	mg/dL forecast

One patient state.
Continuously revised evidence.

Patient history

Phenotype, HbA1c and baseline patterns

Observations

CGM readings, meals, wearable activity

State update

Time-aligned patient context and quality checks

Forecast

30 / 60 / 120-minute glucose scenarios

Clinician view

Trend, contributing context and availability

Safety path: unreliable or stale glucose → withhold prediction. **Feedback path:** a later measured glucose value is compared with the forecast issued earlier; the result becomes part of the audit record.

From observation to forecast on a clinician-facing screen.

144 mg/dL

Valid synthetic glucose reading, 12:15 PM

The dashboard displays the recent glucose trajectory and forecasts, with contributing meal and activity context.

Forecast horizon

30 min → 155.5 mg/dL

60 min → 156.0 mg/dL

120 min → 145.4 mg/dL

The default interactive replay is simulated. An uncalibrated two-hour risk percentage is displayed only as illustrative.

See the original product: run the FastAPI application and select “Guided judge demo.”

Sometimes the safest prediction is no prediction.

12:15 PM · Valid signal

144 mg/dL

Current glucose observed
30 / 60 / 120-minute forecasts shown

2:25 PM · Sensor issue

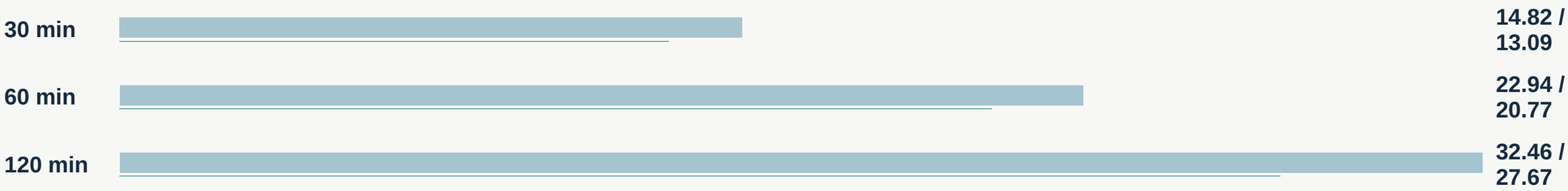
Unavailable

No verified glucose
Forecast and excursion risk withheld

The demonstration includes a 15-minute sensor-freshness rule and rejects missing or unreliable glucose rather than showing a deceptively precise forecast.

Measured improvement over a simple persistence baseline.

Separate CGMacros study · 45 participants, including 14 with Type-2 diabetes · primary patient-grouped holdout: 10 participants (3 with T2D) · lower MAE is better



Bars: gray = persistence MAE, teal = research model MAE; units mg/dL; three forecast horizons. Study reductions: 11.65%, 9.47%, 14.76% respectively.

Scientific boundary: these results are from an independent research experiment, not the model running in the simulated dashboard. Personalized adaptation and trained uncertainty intervals have not passed approval gates. No clinical efficacy or deployment approval is claimed.